Lab 5: Static Code Analysis — Evidence Screenshots

Name: B MEGHANA

SRN: PES2UG24CS103

Environment Setup

```
(.venv) PS D:\L5_37_todo_manager> python --version
>> python -c "import sys; print(sys.executable)"
>> git branch
>> git status
Python 3.14.0
D:\L5_37_todo_manager\.venv\Scripts\python.exe
* lab5-static-analysis
  main
On branch lab5-static-analysis
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   .gitignore

no changes added to commit (use "git add" and/or "git commit -a")
(.venv) PS D:\L5_37_todo_manager>
```

Baseline Analysis ReportS

```
D:\L5_37_todo_manager\.venv\Scripts\python.exe
* lab5-static-analysis
  main
On branch lab5-static-analysis
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   .gitignore    Get-Content .\pylint_report_before.txt | Select-Object -Last
 5
>> Get-Content .\bandit_report_before.txt | Select-Object -Last 12
>> Get-Content .\flake8_report_before.txt
todo_manager.py:60:0: C0116: Missing function or method docstring (missing-function-docstring)

--------------------------------------------------------------------
Your code has been rated at 5.09/10 (previous run: 5.09/10, +0.00)

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 2
                Medium: 1
                High: 1
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 4
Files skipped (0):
todo_manager.py:12:11: E225 missing whitespace around operator
todo_manager.py:40:1: E303 too many blank lines (3)
todo_manager.py:69:49: E261 at least two spaces before inline comment
todo_manager.py:74:38: E231 missing whitespace after ','
(.venv) PS D:\L5_37_todo_manager>
```

Original Program Run

```
(.venv) PS D:\L5_37_todo_manager> python todo_manager.py
No task with id 99
Pending tasks: ['Submit lab report', 'Book train tickets', 'Call plumber']
Task id for 'plumber': 3
Task id for 'gym': None
Try this next: Submit lab report
Tasks restored from file: 4
Exported tasks to tasks.json
(.venv) PS D:\L5_37_todo_manager> python -m pylint todo_manager.py > pylint_report_after.txt 2>&1
>> python -m bandit -r todo_manager.py > bandit_report_after.txt 2>&1
>> python -m flake8 todo_manager.py > flake8_report_after.txt 2>&1
```

Intermediate Static Analysis Results

```
(.venv) PS D:\L5_37_todo_manager> Get-Content .\pylint_report_after.txt
>> Get-Content .\bandit_report_after.txt
>> Get-Content .\flake8_report_after.txt
************* Module todo_manager
todo_manager.py:106:4: C0415: Import outside toplevel (smtplib) (import-outside-toplevel)
todo_manager.py:107:4: C0415: Import outside toplevel (email.message.EmailMessage) (import-outside-toplevel)
todo_manager.py:108:4: C0415: Import outside toplevel (email.utils.parseaddr) (import-outside-toplevel)

-------------------------------------------------------------------
Your code has been rated at 9.67/10 (previous run: 5.09/10, +4.58)

python : [main] INFO    profile include tests: None
At line:2 char:1
+ python -m bandit -r todo_manager.py > bandit_report_after.txt 2>&1
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : NotSpecified: ([main]     INFO    profile inc
   lude tests: None:String) [], RemoteException
    + FullyQualifiedErrorId : NativeCommandError

[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.14.0
Run started:2026-10-14 17:16:06.171998+00:00
```

Bandit Baseline Findings

```
Test results:
>> Issue: [B311:blacklist] Standard pseudo-random generators are not suitable for security/cryptographic purposes.
   Severity: Low   Confidence: High
   CWE: CWE-330 (https://cwe.mitre.org/data/definitions/330.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b311-random
   Location: .\todo_manager.py:138:21
137         if pending:
138             suggestion = random.choice(pending)
139             print("Try this next:", suggestion)

--------------------------------------------------

Code scanned:
        Total lines of code: 115
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 1
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 1
Files skipped (0):
(.venv) PS D:\L5_37_todo_manager>
```

Final Analysis Results

```
(.venv) PS D:\L5_37_todo_manager> python -m pylint todo_manager.py > pylint_report_after.txt 2>&1
(.venv) PS D:\L5_37_todo_manager> cmd /c "python -m bandit -r todo_manager.py > bandit_report_after.txt 2>&1"
(.venv) PS D:\L5_37_todo_manager> python -m flake8 todo_manager.py > flake8_report_after.txt 2>&1
(.venv) PS D:\L5_37_todo_manager> Get-Content .\pylint_report_after.txt
>> Get-Content .\bandit_report_after.txt
>> Get-Content .\flake8_report_after.txt

-------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 9.67/10, +0.33)

[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.14.0
Run started:2026-10-14 17:21:30.352645+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 115
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
```

Final Program Execution

```
(.venv) PS D:\L5_37_todo_manager> python todo_manager.py
No task with id 99
Pending tasks: ['Submit lab report', 'Book train tickets', 'Call plumber']
Task id for 'plumber': 3
Task id for 'gym': None
Try this next: Book train tickets
Tasks restored from file: 4
Exported tasks to tasks.json
```

JSON Task Output

```
(.venv) PS D:\L5_37_todo_manager> Get-Content .\tasks.json
{
  "1": {
    "title": "Submit lab report",
    "priority": "high",
    "labels": [
      "high"
    ],
    "done": false,
    "created": "2026-10-14T22:51:55.549421"
  },
  "2": {
    "title": "Buy groceries",
    "priority": "medium",
    "labels": [
      "medium"
    ],
    "done": true,
    "created": "2026-10-14T22:51:55.549464"
  },
  "3": {
    "title": "Call plumber",
    "priority": "low",
    "labels": [
      "low"
    ],
    "done": false,
    "created": "2026-10-14T22:51:55.549469"
  },
  "4": {
    "title": "Book train tickets",
    "priority": "high",
    "labels": [
      "high"
    ],
    "done": false,
    "created": "2026-10-14T22:51:55.549472"
  }
}
```